

SALES PERFORMANCE, PROFITABILITY AND BUSINESS INTELLIGENCE DASHBOARD USING POWER BI

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Abstract

This project presents a Power BI-based business intelligence solution for retail sales analysis. The implementation consolidates yearly sales data into a Fact_Sales table, applies a star-schema-oriented data model, establishes a dedicated date dimension, and uses DAX measures for sales, profitability, time intelligence, and performance analysis. Interactive dashboard pages were developed for executive overview, sales trends, products, customers, salespersons, geography, budget-versus-actual analysis, and business insights. The submitted dashboard reports approximately \$25.66M in total sales, \$8.34M in gross profit, a 32.52% gross margin, 10,889 orders, 21,078 units, an average order value of about \$2.36K, and 76.58% YoY growth. The report also identifies the limitations of partial 2023 coverage, California-only state representation, Customer ID-based customer visualization, and budget-period alignment.

Keywords— Power BI, Business Intelligence, Sales Analytics, Profitability, DAX, Star Schema, Data Modeling, Data Visualization, Retail Analytics, Dashboard

I. INTRODUCTION

Retail organizations generate transaction data across products, customers, locations, salespersons, and reporting periods. The assignment addressed the difficulty of obtaining a centralized view of sales performance and profitability from manually prepared Excel reports. The developed Power BI solution organizes the supplied data into an analytical model and provides interactive visual analysis for management-oriented questions such as sales growth, product performance, customer contribution, salesperson performance, geographic variation, and budget comparison.

II. PROJECT OBJECTIVES

- Measure revenue, orders, quantity, profitability, margin, and average order value.
- Compare performance across years and analyze daily, monthly, quarterly, and yearly trends.
- Identify high- and low-performing products, customers, salespersons, and locations.
- Evaluate actual sales against available budget information and analyze variance.
- Provide interactive filtering and visual analysis for business decision support.

III. DATASET AND DATA PREPARATION

The supplied Excel workbook was imported into Power BI and inspected using Power Query. Data profiling was used to review data types, validity, blanks, errors, uniqueness, and distributions. The yearly sales tables Sales_2021, Sales_2022, and Sales_2023 were appended into the consolidated Fact_Sales table. The resulting fact table contains Order ID, Product ID, Location ID, Sales Person ID, Customer ID, Order Date, Quantity, and Price. Dimension-oriented tables were retained for product, location, customer, salesperson, and date analysis.

- Imported and inspected the relevant workbook tables.
- Verified identifiers, dates, quantities, and prices.
- Checked data validity, blanks, errors, uniqueness, and distributions.
- Appended the three yearly sales tables into Fact_Sales.
- Prepared dimension tables and validated model relationships.

IV. DATA MODEL AND STAR SCHEMA

The analytical model follows a star-schema-oriented structure in which Fact_Sales stores transaction-level measures and foreign-key-style identifiers, while descriptive dimensions provide filtering and grouping attributes. The implemented relationships connect Dim_Date, Dim_Product, Location_Data, Customer_Data, and Salespeople_Data to Fact_Sales. This structure supports reusable measures and consistent cross-page filtering.

- **Fact table:** Fact_Sales.
- **Dimensions:** Dim_Date, Dim_Product, Location_Data, Customer_Data, Salespeople_Data.
- **Relationships:** each dimension filters Fact_Sales through its corresponding key.

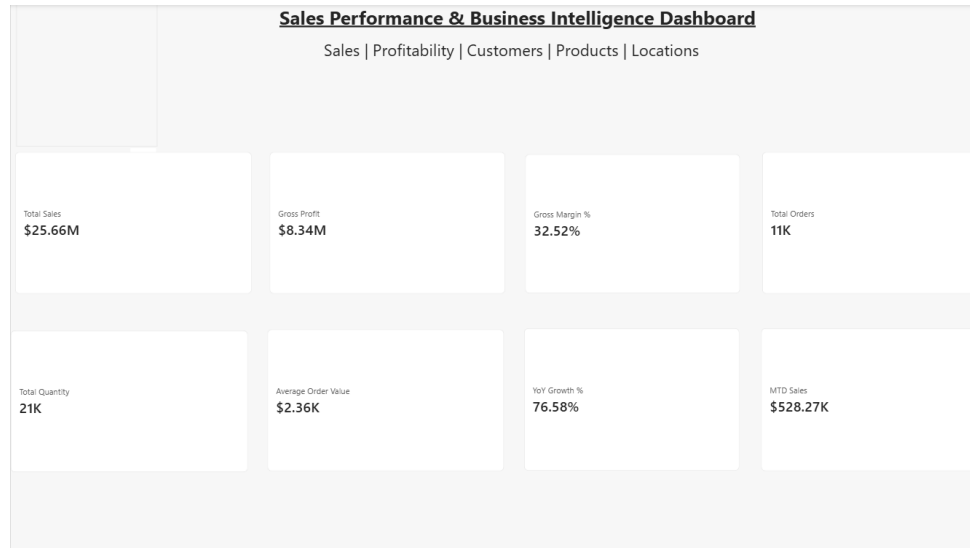


Fig. 1. Executive Overview showing consolidated sales, profitability, orders, quantity, AOV, YoY growth, and MTD sales KPIs.

V. DATE DIMENSION

A dedicated Dim_Date table was created from the transaction date range. It contains Date, Year, Quarter, Month Number, Month Name, Week, Day, and Year-Month. Month Name was sorted using Month Number to preserve chronological ordering. The date dimension supports time-based filtering and DAX calculations such as previous-year comparison, month-to-date, and year-to-date analysis.

- Date
- Year, Quarter, Month Number, Month Name
- Week, Day, Year-Month

VI. DAX MEASURES AND ANALYTICAL CALCULATIONS

Measures were created to provide reusable calculations across dashboard pages. Core measures cover sales, orders, quantity, average order value, average selling price, cost, gross profit, gross margin, revenue per unit, and profit per unit. Time-intelligence measures extend the analysis to previous-year sales, YoY growth, MTD sales, YTD sales, and previous-year YTD sales.

- **Commercial metrics:** Total Sales, Total Orders, Total Quantity, Average Order Value, Average Selling Price.
- **Profitability metrics:** Total Cost, Gross Profit, Gross Margin %, Revenue per Unit, Profit per Unit.
- **Time metrics:** Previous Year Sales, YoY Sales Growth, YoY Growth %, MTD Sales, YTD Sales, Previous Year YTD Sales.

The implementation uses DAX functions including SUMX, AVERAGEX, DIVIDE, CALCULATE, SAMEPERIODLASTYEAR, DATESMTD, and TOTALYTD. These calculations allow the same business logic to respond dynamically to slicers and visual filter context.

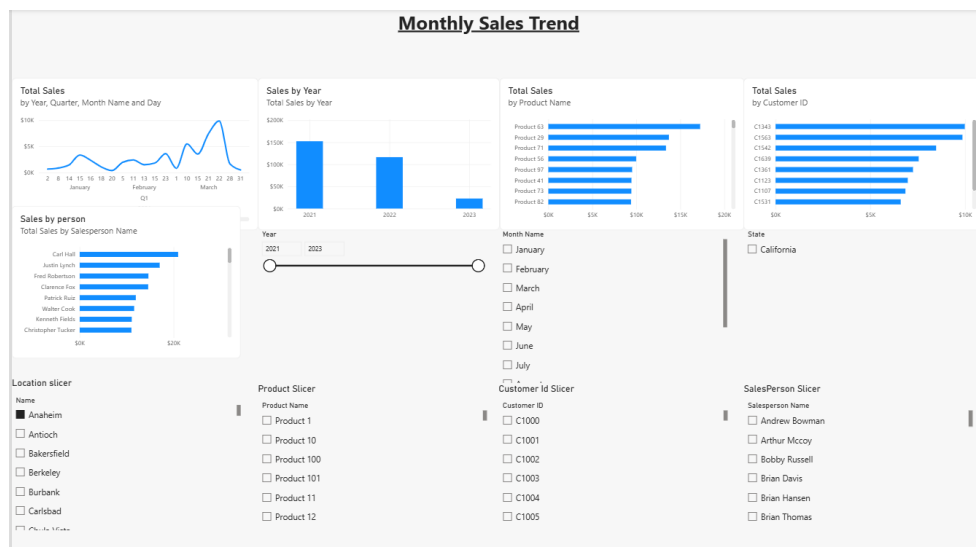


Fig. 2. Monthly Sales Trend page combining daily/monthly trend analysis, yearly comparison, product and customer views, salesperson analysis, and interactive slicers.

VII. DASHBOARD DESIGN AND IMPLEMENTATION

A. Executive Overview

The Executive Overview consolidates the principal KPIs into a single management view. The page presents Total Sales, Gross Profit, Gross Margin, Total Orders, Total Quantity, Average Order Value, YoY Growth, and MTD Sales. It provides a high-level starting point before users move to detailed analytical pages.

B. Monthly Sales Trend

The Monthly Sales Trend page provides time-oriented analysis using daily/monthly trends and yearly comparison. The page also includes product, customer, salesperson, location, state, year, and month filters, enabling interactive exploration of the sales pattern.

C. Product Analysis

Product Analysis evaluates sales, quantity, cost, profit, margin, and average selling price. Top- and bottom-performing products, product contribution, and the relationship between sales and gross profit are presented through bar charts, scatter analysis, and a detailed table.

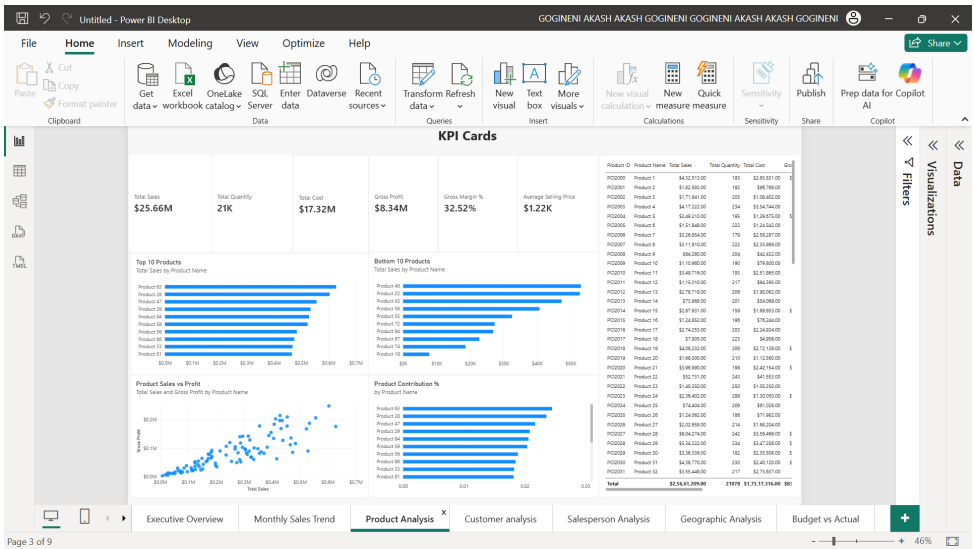


Fig. 3. Product Analysis page showing KPI cards, top and bottom product sales, product contribution, sales-versus-profit scatter analysis, and detailed product metrics.

D. Customer Analysis

Customer Analysis provides a customer-level view of total customers, sales, orders, quantity, and average order value. The submitted dashboard uses Customer ID for the customer visualization, together with a sales-versus-profit scatter plot and a detailed performance table.

E. Salesperson Analysis

Salesperson Analysis compares sales, orders, quantity, gross profit, average order value, and YoY growth across salespersons. A ranking chart, sales-versus-profit scatter plot, and performance table support individual and comparative assessment.

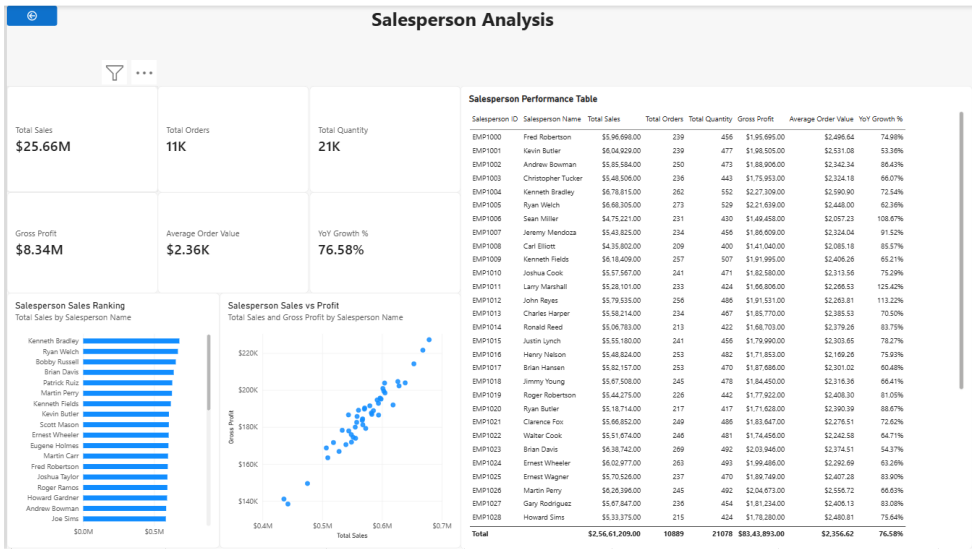


Fig. 4. Salesperson Analysis showing KPI cards, salesperson sales ranking, sales-versus-profit relationship, and salesperson performance table.

F. Geographic Analysis

Geographic Analysis examines sales and profitability by location. The dashboard provides state-level and location-level sales views, location profitability, and a sales-versus-profit scatter analysis. Because the submitted geographic data represents California as the state, the location-level view is more informative than state-to-state comparison within this dataset.

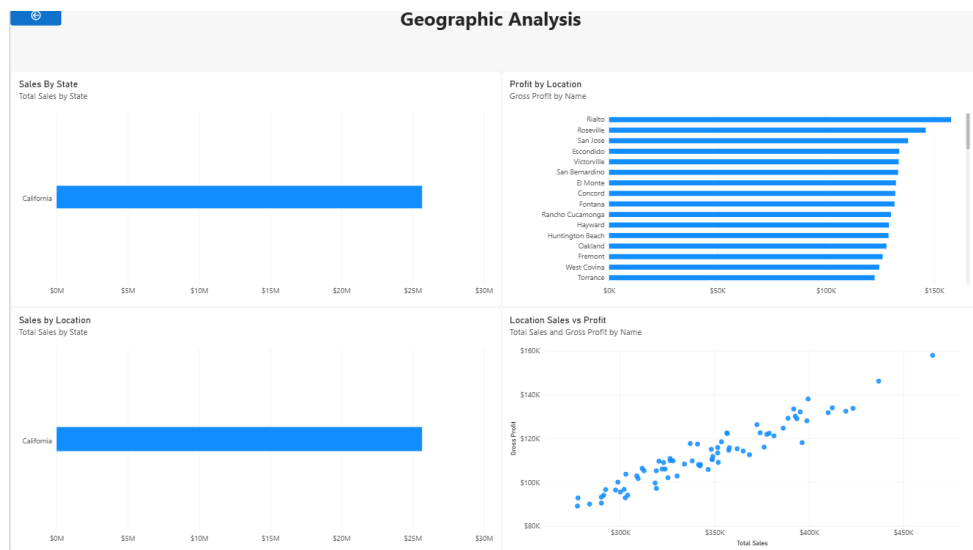


Fig. 5. Geographic Analysis showing state sales, location sales/profitability, and the relationship between location sales and gross profit.

G. Budget vs. Actual

The Budget vs. Actual page compares available budget information with actual sales using budget total, actual sales, variance, variance percentage, and achievement percentage. Monthly and location-level visuals support variance inspection. The interpretation of achievement percentages requires careful period alignment between the budget information and the sales data.

H. Business Insights and Recommendations

The final insights page translates dashboard observations into concise findings, problems, and recommendations. It focuses on overall sales performance, profitability, product variation, location variation, customer variation, and salesperson performance without extending beyond the evidence displayed in the dashboard.

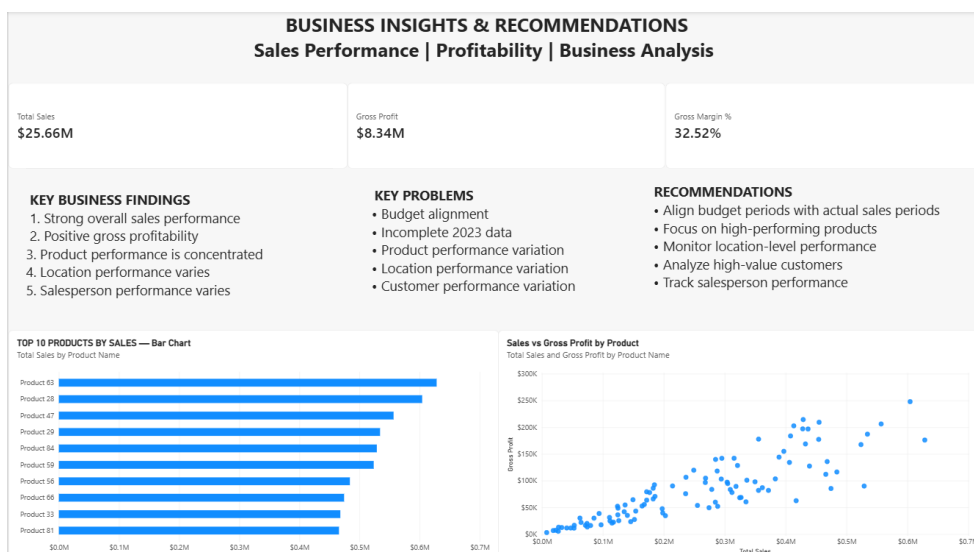


Fig. 6. Business Insights and Recommendations page summarizing the dashboard findings, key problems, recommendations, and product sales-versus-profit visuals.

VIII. RESULTS AND BUSINESS ANALYSIS

The submitted dashboard reports approximately \$25.66M in Total Sales and approximately \$8.34M in Gross Profit, corresponding to a 32.52% Gross Margin. The report also shows 10,889 Total Orders and 21,078 units sold. The Average Order Value is approximately \$2.36K, while the displayed YoY Growth is approximately 76.58%.

The product analysis demonstrates variation across products and provides both top- and bottom-oriented views. The scatter analysis shows the relationship between sales and gross profit across products. Customer and salesperson pages similarly expose performance variation through rankings, profitability comparisons, and detailed tables. Geographic analysis shows substantial location-level variation while state-level analysis is constrained by the available state coverage.

The Budget vs. Actual page provides an additional management perspective by comparing actual sales with the available budget structure. However, budget achievement should not be treated as a direct annual performance measure until the budget and actual periods are aligned.

IX. KEY BUSINESS FINDINGS

- Overall sales performance is substantial, with approximately \$25.66M in reported sales.

- The dashboard reports approximately \$8.34M gross profit and a 32.52% gross margin.
- Product performance varies materially, with clear differences between higher- and lower-performing products.
- Customer performance varies across sales, orders, quantity, average order value, and profitability.
- Salesperson performance varies across sales, profit, orders, and YoY growth.
- Location-level performance varies, while state-level comparison is limited by the available California-only state representation.
- Budget-versus-actual analysis requires careful interpretation because the reporting periods are not directly equivalent.

X. KEY PROBLEMS IDENTIFIED

- **Budget alignment:** budget and actual periods need consistent time alignment before achievement percentages are interpreted as direct annual comparisons.
- **Incomplete 2023 data:** the 2023 sales period is partial compared with earlier years, limiting full-year comparison.
- **Product performance variation:** product contribution is uneven and requires continued monitoring.
- **Location performance variation:** location-level sales and profit differ across the available locations.
- **Customer performance variation:** customer-level contribution differs across sales, orders, AOV, and profitability.

XI. RECOMMENDATIONS

- Align budget periods with the corresponding actual sales periods before performance comparison.
- Monitor high-performing products and investigate lower-performing products using sales and profitability measures.
- Monitor location-level performance to identify areas requiring attention.
- Analyze high-value customers using sales, orders, AOV, and profitability together.
- Track salesperson performance using sales, profit, orders, and YoY growth.

XII. LIMITATIONS

- The 2023 sales data is incomplete/partial compared with earlier years.
- The geographic data represents California as the state, limiting state-to-state comparison.
- Customer analysis in the submitted dashboard is displayed using Customer ID rather than customer name.
- Budget and actual periods require alignment before achievement percentages can be interpreted as a direct annual comparison.
- The report does not claim interactive features or analytical fields that are not demonstrated in the supplied material.

XIII. CONCLUSION

The implemented Power BI solution provides a centralized analytical environment for sales monitoring, profitability analysis, product analysis, customer analysis, salesperson analysis, geographic analysis, and budget comparison. The star-schema-oriented model, dedicated date dimension, reusable DAX measures, and interactive dashboard pages transform the supplied transaction data into a structured business intelligence workflow. The reported results demonstrate meaningful visibility into sales and profitability, while the documented limitations provide appropriate context for interpreting year-over-year and budget comparisons.

REFERENCES

- [1] Power BI Business Intelligence Project: Sales Performance, Profitability & Business Intelligence Dashboard, Assignment Brief.
- [2] Source retail sales dataset/workbook supplied with the assignment.
- [3] Microsoft Power BI and DAX documentation, used as technical reference for Power BI modeling, measures, and time-intelligence calculations.